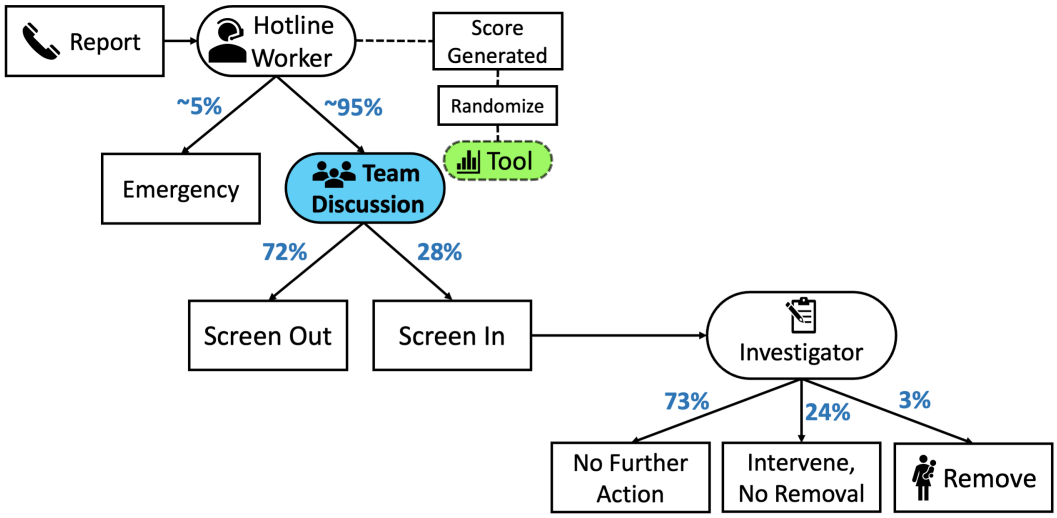
Online Appendix

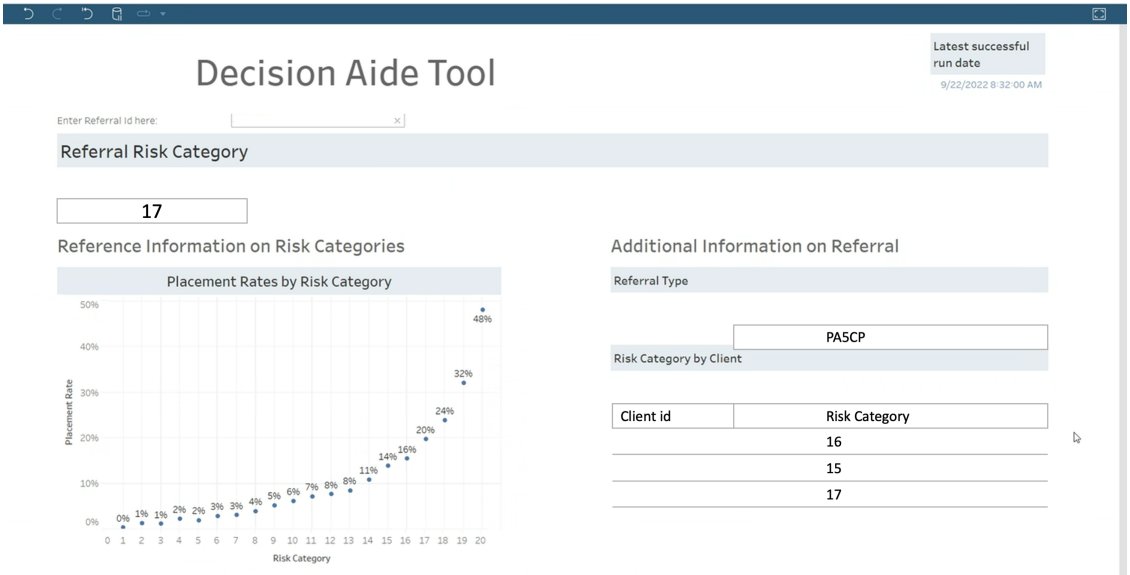
Appendix A

Figure A.1: Description of Child Protective Services in the Evaluation County



Notes: Listed percentages are the share of children in the county assigned to the respective decision, conditional on reaching the given node. For example, three percent of investigated children are placed in foster care (removed from home) within 90 days.

Figure A.2: Algorithm Tool Display for Workers

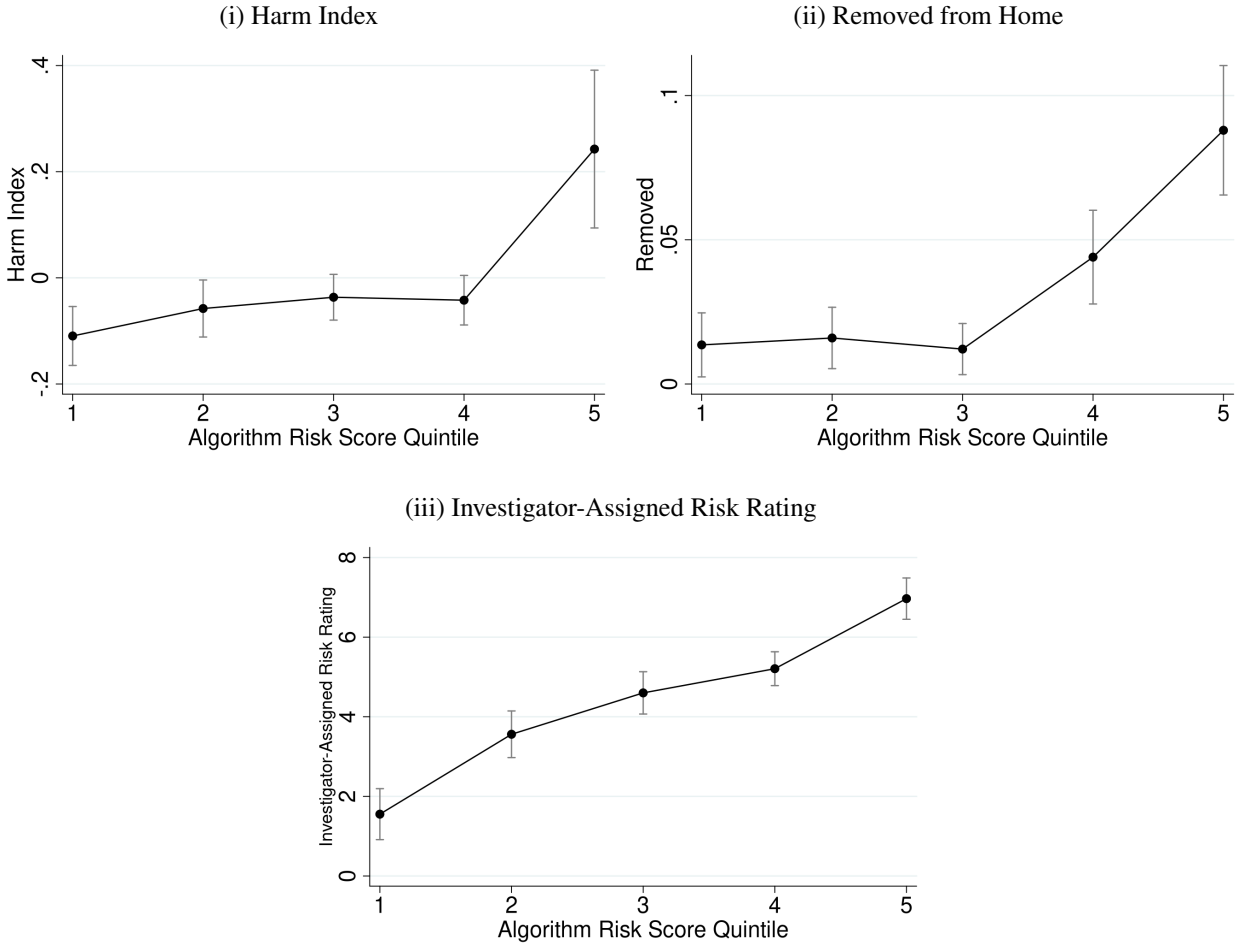


Notes: This figure shows an example of the algorithmic tool interface. The right panel shows predicted risk scores for all children listed on the referral (in this case 16, 15, and 17). In the top left of the display, the highest score across all of the children is

emphasized to decision makers (in this case 17). The plot on the bottom left reminds workers of the chance that a child is removed from their home (foster care placement) within two years at each score level.

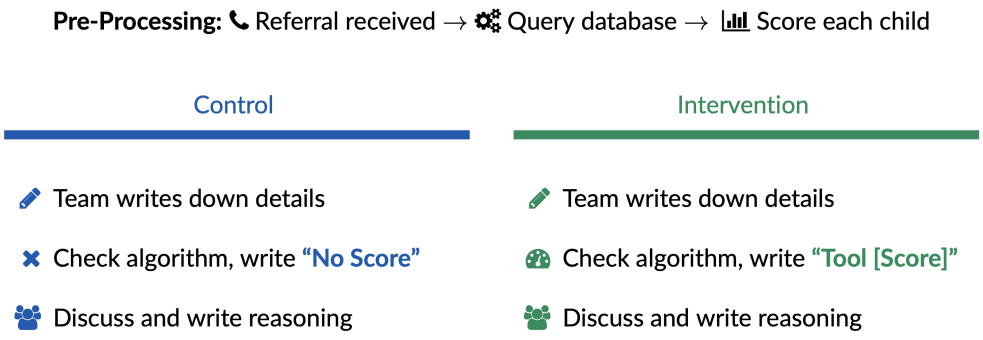
Figure A.3: Algorithm Risk Score Predicts Relevant Outcomes

Control Group



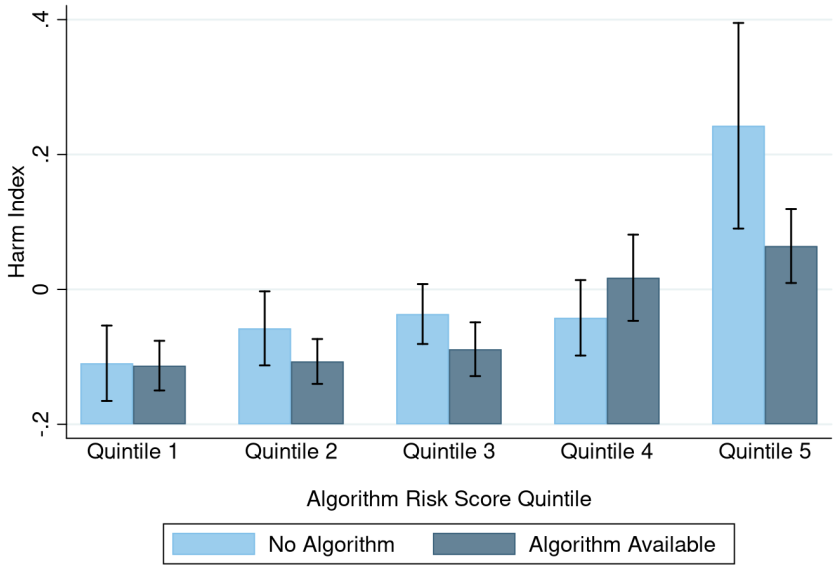
Notes: These subfigures show the relationship between relevant outcomes and the algorithm risk score, binned by quintile for visual clarity. Each point represents the mean of the outcome variable in the control group at a given quintile of predicted risk, with a 90% confidence interval reported. The algorithm was trained on prior data to predict a child's removal from their home to foster care within two years of a referral. Investigator-assigned risk ratings (Panel iii) come from visits to the home and are only available for the set of screened-in children.

Figure A.4: Trial Workflow



Notes: This figure presents a stylized summary of teams' decision workflow during the trial. Algorithm scores were available for the intervention group, and unavailable for the control group. Workers were encouraged, but not required, to check the algorithm and write down "No Score" if the score was unavailable or "Tool [Score]" with the specific listed score(s).

Figure A.5: Impact of Algorithm on Child Harm, by Predicted Risk Level



Notes: This figure presents mean child harm by algorithm score quintile, with and without the tool available, with 90% confidence intervals. The sample is restricted to children randomized during the first year of the trial to allow for several months of potential hospitalization following randomization. The harm index is constructed as a standardized sum of standardized outcomes including a child's number of high-priority hospital admissions, number of admissions with a listed injury, number of avoidable ER visits, substance exposure incidents, number of admissions with a confirmed maltreatment code, and number of visits with an intentional injury (assault, self-harm) code. The index is standardized (mean 0, variance 1) on the control group and a lower value is considered better.